

HK IMO Training 2020-2021
Problem Set 1
For 10 October 2020



Problem 1. Two circles ω_1 and ω_2 are externally tangent to each other. A common external tangent of these two circles touches ω_1 at A and ω_2 at B . Segment AP is the diameter of the circle ω_1 , and the line passing through point P is tangent to ω_2 at point Q .

Prove that $AP = PQ$.

Problem 2. Let Γ_1 and Γ_2 be two non-overlapping circles. Suppose

- Points A, C are on Γ_1 and points B, D are on Γ_2 such that AB is a common external tangent to the two circles, and CD is a common internal tangent to the two circles,
- AC and BD meet at E ,
- F is a point on Γ_1 ,
- the tangent line to Γ_1 at F meets the perpendicular bisector of EF at M , and
- MG is a line tangent to Γ_2 at G .

Prove that $MF = MG$.

Problem 3. In $\triangle ABC$ with $AB \neq AC$, points X and Y are projections of B and C , respectively, to the internal angle bisector of $\angle A$. Let S and T be distinct points on side BC such that $\angle XSY = \angle XTY = 90^\circ$. Show that the circumcircles of $\triangle AST$ and $\triangle ABC$ are tangent to each other.

Problem 4. Let $\triangle ABC$ be an acute triangle with $AB < AC$ and circumcentre O . Let Q be the intersection of the external bisector of $\angle A$ with BC , and let P be a point in the interior of $\triangle ABC$ such that $\triangle BPA$ is similar to $\triangle APC$.

Show that $\angle QPA + \angle OQB = 90^\circ$.